

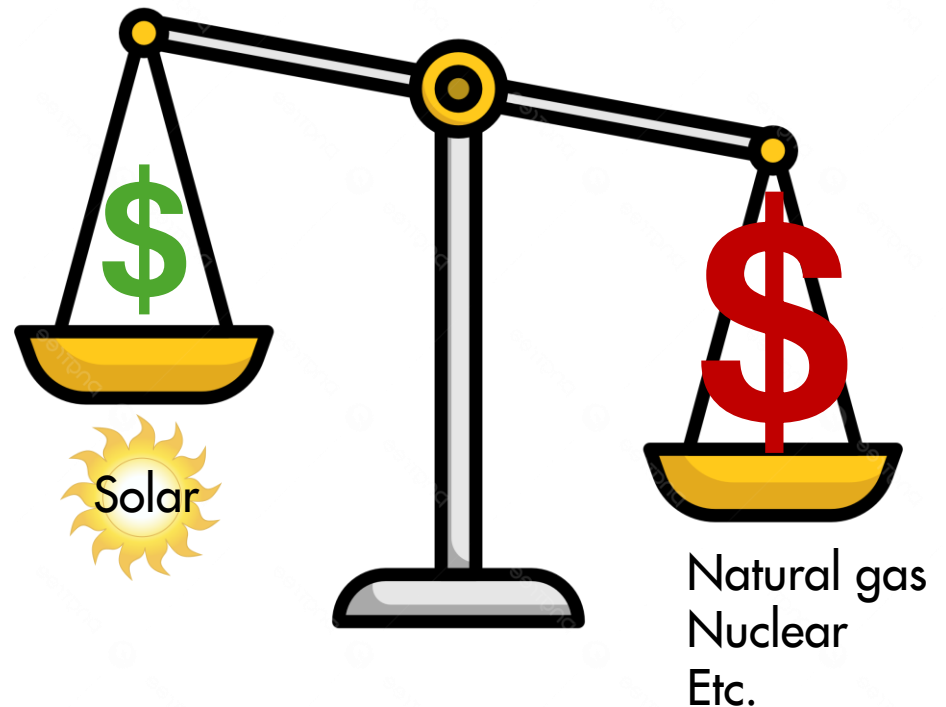
Predicting Day-Ahead Solar Energy Production

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Special thanks to: Alex Myers, Praveen Shahani, Steven Gubkin

Why (study) solar?

- Low marginal cost
- Abundant resource availability
- Slow degradation





Energy
Market



How does the energy market work?





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- Daily bidding!
- Bid TODAY by 10am for all of TOMORROW'S energy

I'll provide X
mWh for \$Y at
time T tomorrow.



I'll provide A
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I'll buy all the
CHEAPEST first,
then go more
expensive.



**Goal: (Skip a) day-
ahead prediction of
hour-by-hour energy
production**



The data

Solar data:

- NLR PVDAQ
(National Laboratory of the Rockies)

Weather data:

- Open Meteo



The data: difficulties

- Inconsistency between systems
- Inconsistency within a system
- Downtime/bad readings



The data: ~~difficulties~~ solutions

- Inconsistency between systems
 - Inconsistency within a system
 - Downtime/bad readings
- } Flag using EDA + investigate outliers by hand

Restricted days we considered. Required:

- **Frequent** power readings
- **Sufficient number** of power readings



The target: What do outputs look like?

If today is
1/1/2026...

Time (hourly)	Energy produced (kWh)
1/2/2026 at 8:00	???
1/2/2026 at 9:00	???
1/2/2026 at 10:00	???
1/2/2026 at 11:00	???
1/2/2026 at 12:00	???
1/2/2026 at 13:00	???
1/2/2026 at 14:00	???
1/2/2026 at 15:00	???
1/2/2026 at 16:00	???
1/2/2026 at 17:00	???



The features: What do inputs look like?

Goal: Predict energy on 1/2/2026

Inputs: Data from 12/31/2025 and earlier*.

For each day, some subset of:

- Hourly energy production
- Some past energy production ("lags")
- Hourly cloud cover
- Hourly irradiance
- Proportion daytime

*actual (as a proxy for historical forecasts)



**So how did we predict
and how well did we
predict?**



The judge

- Asymmetric mean squared error

$$\text{error} = \frac{a \cdot (\text{underestimate})^2 + b \cdot (\text{overestimate})^2}{\# \text{ observations}}$$

(we used $a = 1$, $b = 2$; in reality should depend on cost-profit tradeoff)



Top-performing model (mean errors)

	System 10	System 50	System 51
Recorded max energy (kWh)	1.186	7.073	7.237
Naïve (baseline)	0.0445978	1.747726	1.4837508
Prophet	0.029864	1.253558	0.9606655
LinReg	0.033711	1.198927	1.102705
SARIMAX*	0.0254795	0.4169801	0.570796
Light GBM	0.039082	1.019386	1.099389
XGBoost	0.039193	1.041629	1.098982
Best performing model's error on test set	0.023272	0.312543	0.413511



Summary:

- **Solar energy**
- **Customized errors**
- **SARIMAX**

